Operating

System Security

(MITS 74915)

Submitted to,

Name: Julie Thorpe

Faculty of Business and Information Tech

Submitted by,

Name: Avishkar Sharma

Assignment: 2

Question Number: 1

1.a

```
Adm_AvishkarSharma@fedora:~$ ./AvishkarSharma_app &
[1] 9254
param1=value1
param2=value2
param3=value3
param4=value4
param5=value5
You have a large params file, that's good to start with for testing...
Adm_AvishkarSharma@fedora:~$ ps -axZ | grep AvishkarSharma_t
unconfined_u:unconfined_r:AvishkarSharma_t:s0-s0:c0.c1023 9254 pts/0 S    0:00 ./AvishkarSharma_app
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 10069 pts/0 S+   0:00 grep --color=auto AvishkarSharma_t
Adm_AvishkarSharma@fedora:~$
```

Figure 1: Output Showing the application running in the AvishkarSharma_t domain

1.b

```
Adm_AvishkarSharma@fedora:~$ ls -Z AvishkarSharma_app
unconfined_u:object_r:user_home_t:s0 AvishkarSharma_app
Adm_AvishkarSharma@fedora:~$ ls -Z priv/
system_u:object_r:user_home_t:s0 params.txt  system_u:object_r:user_home_t:s0 records.txt
Adm_AvishkarSharma@fedora:~$ make -f /usr/share/selinux/devel/Makefile AvishkarSharma_myFirstPolicy.pp
/usr/share/selinux/devel/include/distributed/smartmon.if:13: Warning: duplicate definition of smartmon_read_tmp_files(). Original definition on /usr/share/selinux/devel/include/contrib/smart
mon.if:13.
/usr/share/selinux/devel/include/distributed/smartmon.if:39: Warning: duplicate definition of smartmon_admin(). Original definition on /usr/share/selinux/devel/include/contrib/smartmon.if:39
.
Compiling targeted AvishkarSharma_myFirstPolicy module
Creating targeted AvishkarSharma_myFirstPolicy.pp policy package
rm tmp/AvishkarSharma_myFirstPolicy.mod tmp/AvishkarSharma_myFirstPolicy.mod.fc
Adm_AvishkarSharma@fedora:~$ sudo semodule -i AvishkarSharma_myFirstPolicy.pp
Adm_AvishkarSharma@fedora:~$ sudo semanage fcontext -a -t AvishkarSharma_data_t "/home/Adm_AvishkarSharma/priv(/.*)?"
File context for /home/Adm_AvishkarSharma/priv(/.*)? already defined, modifying instead
Adm_AvishkarSharma@fedora:~$ sudo restorecon -vRF /home/Adm_AvishkarSharma/priv
Relabeled /home/Adm_AvishkarSharma/priv from system_u:object_r:user_home_t:s0 to system_u:object_r:AvishkarSharma_data_t:s0
Relabeled /home/Adm_AvishkarSharma/priv/records.txt from system_u:object_r:user_home_t:s0 to system_u:object_r:AvishkarSharma_data_t:s0
Relabeled /home/Adm_AvishkarSharma/priv/params.txt from system_u:object_r:user_home_t:s0 to system_u:object_r:AvishkarSharma_data_t:s0
Adm_AvishkarSharma@fedora:~$ sudo restorecon -v AvishkarSharma_app
Relabeled /home/Adm_AvishkarSharma/AvishkarSharma_app from unconfined_u:object_r:user_home_t:s0 to unconfined_u:object_r:AvishkarSharma_exec_t:s0
Adm_AvishkarSharma@fedora:~$ ls -Z AvishkarSharma_app
unconfined_u:object_r:AvishkarSharma_exec_t:s0 AvishkarSharma_app
Adm_AvishkarSharma@fedora:~$ ls -Z priv/
system_u:object_r:AvishkarSharma_data_t:s0 params.txt
system_u:object_r:AvishkarSharma_data_t:s0 records.txt
Adm_AvishkarSharma@fedora:~$ sudo semodule -l | grep AvishkarSharma
AvishkarSharma_myFirstPolicy
Adm_AvishkarSharma@fedora:~$
```

Figure 2: Output showing the module is loaded

1.c

```
Adm_AvishkarSharma@fedora:~$ ./AvishkarSharma_app &
[1] 14876
param1=value1
param2=value2
param3=value3
param4=value4
param5=value5
You have a large params file, that's good to start with for testing...
Adm_AvishkarSharma@fedora:~$ ps axZ | grep AvishkarSharma_t
unconfined_u:unconfined_r:AvishkarSharma_t:s0-s0:c0.c1023 14876 pts/0 S    0:00 ./AvishkarSharma_app
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 14889 pts/0 S+   0:00 grep --color=auto AvishkarSharma_t
Adm_AvishkarSharma@fedora:~$ ls -Z AvishkarSharma_app
unconfined_u:object_r:AvishkarSharma_exec_t:s0 AvishkarSharma_app
Adm_AvishkarSharma@fedora:~$ ls -dZ priv/
system_u:object_r:AvishkarSharma_data_t:s0 priv/
Adm_AvishkarSharma@fedora:~$ ls -Z priv/
system_u:object_r:AvishkarSharma_data_t:s0 params.txt
system_u:object_r:AvishkarSharma_data_t:s0 records.txt
Adm_AvishkarSharma@fedora:~$
```

Figure 3: Output showing the binary and priv directory are labeled correctly.

Question Number: 2

```
policy_module(AvishkarSharma_myFirstPolicy, 1.0)

#1.Declaration and Requirements
#Requirement Block: It will tell the kernel where the existing base types are required to interact with other system
gen_require(`
    type unconfined_t;
    role unconfined_r;
    attribute file_type;
')
# Define the domain for the application process
type AvishkarSharma_t;
# Define the label for executable binary
type AvishkarSharma_exec_t, file_type;
# Define the label for the 'priv' directory and its logs
type AvishkarSharma_data_t, file_type;

#2.Domain Transition Logic
# When a process is unconfined_t executes a file labeled as 'AvishkarSharma_exec_t', new process will automatically transition to 'AvishkarSharma_t' domain
type_transition unconfined_t AvishkarSharma_exec_t:process AvishkarSharma_t;

#3.Execution & Process Permissions
# Allowing the domain 'AvishkarSharma_t' to use the binary as an entrypoint to start the process
allow AvishkarSharma_t AvishkarSharma_exec_t:file { entrypoint read map execute open getattr };
# Allowing the user's unconfined_t to execute the labeled binary
allow unconfined_t AvishkarSharma_exec_t:file { read execute open getattr map };
#Allowing the user's shell to perform the transition to the new domain and receive signals when the application completes its execution
allow unconfined_t AvishkarSharma_t:process { transition sigchld };

#4.Data Storage and File Access
# This allows the app process to enter, search and list the directory, allow 'add_name' and 'write' will allow the process to create new files in the directory
allow AvishkarSharma_t AvishkarSharma_data_t:dir { read search open add_name write };
# Allow the app to read and write the params.txt and records.txt
allow AvishkarSharma_t AvishkarSharma_data_t:file { read write open getattr append create };
```

Figure 4: File Screenshot of AvishkarSharma_myFirstPolicy.te

```
#1.Labeling the Application Binary
#Rule to match the path of executable, "--" specifies rule will be only applied to regular files and assigns 'AvishkarSharma_t' so binary can act as a domain entrypoint
/home/Adm_AvishkarSharma/AvishkarSharma_app  -- gen_context(system_u:object_r:AvishkarSharma_exec_t,s0)

#2.Labeling the private data Directory and Contents
#The Regular Expression (/.*) ensures the 'priv' directory and everything inside is created and also gets the same label, this allows the app to manage its own files
/home/Adm_AvishkarSharma/priv(/.*)? gen_context(system_u:object_r:AvishkarSharma_data_t,s0)
```

Figure 5: File Screenshot of AvishkarSharma_myFirstPolicy.fc

Description of whole scenario:

The policy uses *Type Enforcement* to create a restricted domain name: *AvishkarSharma_t*. Using *Type Transition*, the application is confined to only accessing file labeled with *AvishkarSharma_data_t*, implementing the *Principle of Least privilege*.